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# Q16 - Sampling and Quantization Demonstration

import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Upload image
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read image
img = cv2.imread(filename)
img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

# -----
# Sampling (Reduce Resolution)
# -----
sampled = cv2.resize(img, (img.shape[1]//2, img.shape[0]//2))

# -----
# Quantization (Reduce Color Levels)
# -----
levels = 16 # Number of intensity levels

quantized = np.floor(img / (256 / levels)) * (256 / levels)
quantized = quantized.astype(np.uint8)

# -----
# Display
# -----
plt.figure(figsize=(15,5))

plt.subplot(1,3,1)
plt.imshow(img)
plt.title("Original Image")
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(sampled)
plt.title("After Sampling")
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(quantized)
plt.title("After Quantization")
plt.axis('off')

plt.show()

# -----
# Information
# -----
print("Original Resolution :", img.shape[:2])
print("Sampled Resolution  :", sampled.shape[:2])

original_size = img.nbytes / 1024
sampled_size = sampled.nbytes / 1024

print(f"\nApprox Memory Usage")
print(f"Original : {original_size:.2f} KB")
print(f"Sampled  : {sampled_size:.2f} KB")

print("\nObservation:")
print("- Sampling reduces image resolution, decreasing storage and computation.")
print("- Quantization reduces intensity levels while preserving most visual quality.")
print("- Together they optimize performance with minimal quality loss.")

```

Choose files images (3).jpg

images (3).jpg(image/jpeg) - 44856 bytes, last modified: 21/07/2026 - 100% done

Saving images (3).jpg to images (3).jpg

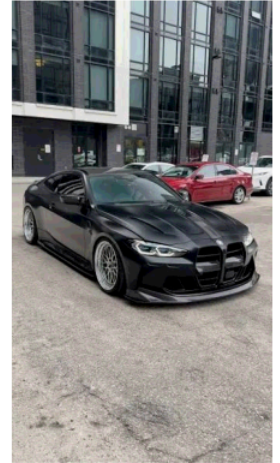
Original Image



After Sampling



After Quantization



Original Resolution : (597, 335)

Sampled Resolution : (298, 167)

Approx Memory Usage

Original : 585.92 KB

Sampled : 145.80 KB

Observation:

- Sampling reduces image resolution, decreasing storage and computation.
- Quantization reduces intensity levels while preserving most visual quality.
- Together they optimize performance with minimal quality loss.